



Coral bleaching simulation

Tutorial for teachers

1. Controls and display

Macroscopic-scale simulation

This simulation allows us to visualise the bleaching of a coral reef comprising three genera of coral – *Acropora*, *Pocillopora* and *Porites* – by varying the ocean temperature over time. These three genera are found in the Pacific Ocean; they do not always coexist, but it is possible for them to do so. Scientists have focused particularly on these three because they exhibit different physiological responses to changes in ocean temperature [1].

The reef depicted therefore comprises several specimens of each of these three genera, which exhibit different levels of sensitivity to rising temperatures. When the temperature and duration of exposure exceed the thresholds specific to each genus, the corals bleach. If the rise persists for too long, this state of distress becomes irreversible and the corals die. The following are the threshold values used in the simulation for each genus:

- **Acropora**, the most sensitive: exposure to temperatures of 31°C or higher for seven consecutive days causes bleaching. After 14 consecutive days, the corals begin to die.
- **Pocillopora**, slightly more resilient: exposure to temperatures of 31°C or higher for 10 consecutive days causes bleaching. After 14 consecutive days, the corals begin to die.
- **Porite**, the most resilient: exposure to temperatures of 32°C or higher for 21 days causes bleaching. After 28 consecutive days, the corals begin to die.

For each group of corals, the simulation does not account for variability in individual resistance, the aim being to be able to compare the groups with one another (see Section 5). The bleaching process begins when the corals start to turn pale. Recovery is possible if the temperature falls below the critical exposure threshold. Once the mortality threshold is reached (in terms of the number of days), recovery is no longer possible; once dead, the corals remain white.

How to read the graph

The graph shows how water temperature (in °C) changes over time (in days).

- The button   starts or pauses the simulation.
- The button  resets the time to zero.
- The greyed-out area of the graph shows how water temperature changes over time. The red dots on the graph indicate the point at which the corals die, with no possibility of recovery. The animation shows only two such points, as corals of the same genus all die simultaneously. It should be noted that both *Pocillopora* and *Acropora* die after the same duration of exposure to 31°C; it is therefore important to distinguish their physiological differences before this common mortality threshold.

- When you click on a red dot, a box shows which genus dies at that time.
- The slider next to the thermometer allows you to adjust the water temperature from below 22 °C to above 36 °C. The current temperature is displayed above the thermometer.
- The Settings button  allows you to switch to the microscopic-scale simulation; show or hide the names of the different genera of corals.
- Clicking on the magnifying glass  located on an Acropora coral also switches to the microscopic view; in microscopic view, click on the coral icon in the bottom right-hand corner to return to the macroscopic view. 

Microscopic-scale simulation

This simulation allows students to visualise the bleaching of a model polyp for the three genera. The threshold values used here are those for the genus *Acropora* (bleaching = 7 days at 31°C, mortality = 14 days at 31°C). This enables students to identify the genus. In this simulation, they are guided to understand the biological mechanism of coral bleaching in general. A single polyp from the reef is shown in magnified form, as in Fig. 1. The base of the polyp is not visible as it is anchored in the reef's calcareous skeleton, but its tentacles extend outwards. It is shown in a frontal cross-section to reveal the zooxanthellae (microscopic algae present in the polyp's external tissues; see Fig. 1 and Box 1).

When the temperature and exposure time exceed the bleaching thresholds, the polyp – now weakened – expels most of its zooxanthellae: deprived of the nutrients produced by the zooxanthellae's photosynthesis, it gradually bleaches. If the rise in temperature persists for too long (mortality threshold exceeded), this state of distress becomes irreversible and the polyp dies and disappears (as its tissue is transparent, see Box 1). All that remains is the calcareous skeleton. Conversely, if temperatures fall below the threshold before 14 days have elapsed, the symbiosis with the zooxanthellae is gradually re-established.

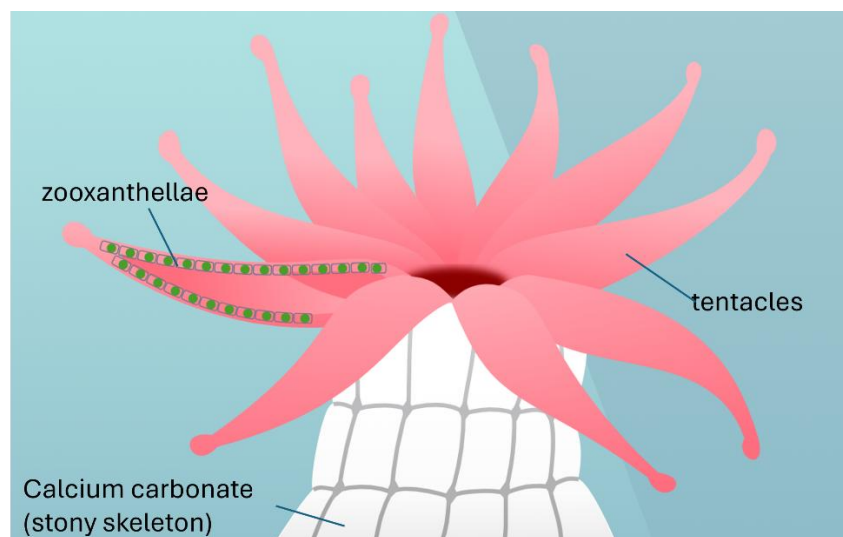
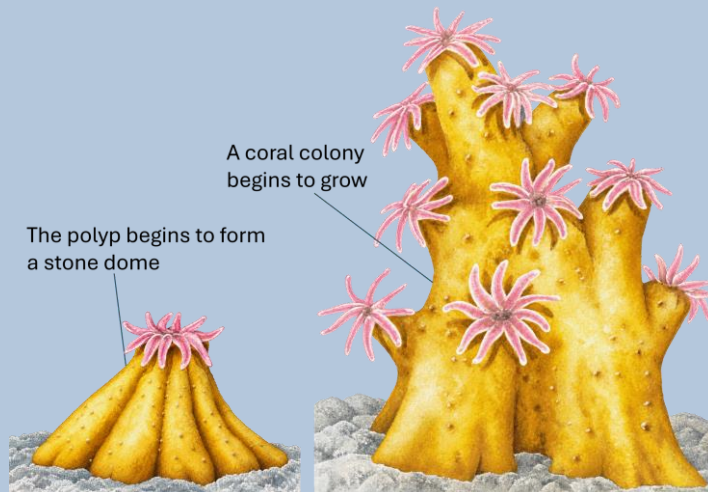


Fig. 1 : Extract from the microscopic-scale simulation

Box 1. Coral anatomy

A polyp is a small, soft-bodied, tubular marine invertebrate which forms the basic unit of coral and secretes a calcareous skeleton within which it lives [2]. The polyp builds its calcareous skeleton by absorbing calcium and carbonate ions from seawater. These ions are transported to the base of the polyp, where they combine to form calcium carbonate. This gradually deposits beneath the polyp and forms a rigid calcareous layer.



As the polyp grows, it continues to deposit calcium carbonate, which allows the skeleton to thicken and the reef to develop (see Fig. 2).

Fig. 2 : The development of a coral reef. Image adapted from 'Coral Reefs'. 2022. Piccolia

The polyp harbours zooxanthellae (microscopic single-celled algae) within its tissues; through photosynthesis, these provide the polyp with most of its nutrients, whilst the polyp, in return, offers them a refuge (see Fig. 3).

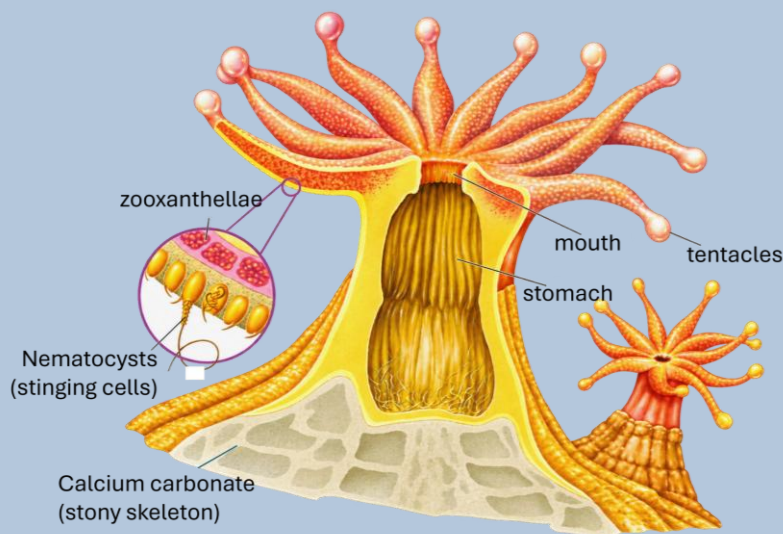


Fig. 3 : Cross-section diagram of a polyp. Image adapted from 'Coral Reefs'. 2022. Piccolia

Normally, zooxanthellae use solar energy for photosynthesis. When the water becomes too warm, they can no longer utilise this energy efficiently and produce substances that can damage both the zooxanthellae and their host polyp. To survive thermal stress, the polyp then expels a large proportion of the zooxanthellae [3]. As the polyp's tissue is transparent, its white skeleton becomes visible: the coral then appears pale or white. This phenomenon is known as coral bleaching [4].

2. About the simulation

This simulation offers students the opportunity to understand the impact of global warming on coral reefs at both the microscopic and macroscopic levels. It enables coral reefs to be explored as a rich biological model, illustrating the mutualistic symbiotic relationship between polyps and zooxanthellae. Reefs are, moreover, extremely diverse ecosystems, harbouring not only numerous species of coral but also a wide variety of other interconnected marine organisms (see Box 2).

Prerequisites

The following concepts are prerequisites for this simulation:

- **Symbiosis:** a long-term relationship between two organisms of different species, in this case of a mutualistic nature, i.e. beneficial to both partners (the polyp and the zooxanthellae).
- **Micro-organism:** a living organism of microscopic size, invisible to the naked eye; illustrated here by zooxanthellae, single-celled algae living within the polyp's tissues.
- **Ecosystem:** the totality of living organisms in a given habitat (in this case, the coral reef) and the interactions they have with one another and with their environment.
- **Photosynthesis:** the process by which an organism (in this case, zooxanthellae) uses light to produce organic matter.
- **Species and genus:** concepts of biological classification, which are useful for understanding why the simulation distinguishes between corals at genus level (Acropora, Pocillopora, Porites) rather than at species level.
- **Global warming:** gradual rise in average global temperatures, of which the rise in ocean temperatures is a direct consequence.

Relationship with the other simulations

This simulation can be supplemented by our other resources in chemistry and physics, as part of a multidisciplinary approach.

It is useful for students to explore the [Ocean Acidification simulation](#) and the [Ocean Acidification activity](#) in their chemistry lessons, in order to distinguish between the phenomena of acidification and bleaching (see Sect. 6).

Furthermore, the various physics simulations and the [activity Understanding the greenhouse effect](#) enable students to understand the causes of global warming, and consequently the rise in sea temperatures.

3. Underlying misconceptions

- Students may associate bleaching with the immediate death of the coral. However, bleaching refers to the loss of symbiotic algae; the coral remains alive, but is severely weakened and more vulnerable to death [4, 5].
- Students may assume that all corals react in the same way to rising temperatures. In reality, sensitivity to thermal stress varies considerably between species and may even differ between individuals of the same species, which justifies a differentiated approach to modelling corals [1, 6].
- Students may think that a reef recovers quickly after a bleaching event. Research shows, however, that recovery is a slow process, often taking several decades, and that repeated heatwaves prevent the reef from returning to its original state [7].
- Ocean acidification and coral bleaching are two distinct phenomena [2], the former being caused by the diffusion of CO₂ molecules from the atmosphere into the ocean (see the tutorial for the 'Ocean Acidification' simulation). However, without studying these phenomena separately in chemistry and biology, students are likely to confuse them, or even attribute bleaching to acidification. Although acidification may be a stress factor that weakens the coral skeleton and is likely to accelerate bleaching (among other factors [5]), the primary cause of bleaching linked to global warming is the rise in water temperature [2].

4. For students to discover

This simulation gives students the opportunity to understand, through active exploration of the two viewing scales, one of the consequences of global warming on coral reefs.

At the macroscopic level, students will be able to observe the different sensitivities and responses to thermal stress exhibited by the three genera of coral (*Acropora*, *Pocillopora* and *Porites*), by comparing their respective bleaching and mortality thresholds. For example, *Pocillopora* and *Acropora* die after the same number of days at 31°C, whilst their bleaching begins at different times, indicating that these two genera have distinct physiologies despite dying at the same time. These observations highlight coral biodiversity by showing that several genera, coexisting on the same reef, do not react in the same way to the same warming episode.

They will also observe that bleaching is, initially, a reversible phenomenon: as long as the mortality threshold specific to each genus has not been reached, a drop in temperature allows the corals to regain their colour. This reversibility is, however, limited in time: once the mortality threshold specific to each genus has been exceeded, the corals' state of distress becomes irreversible and they die.

Through the microscopic view, students will be able to examine, in particular, the mutualistic symbiotic relationship between the polyp and the zooxanthellae (Box 1).

They will be able to observe the bleaching process and thereby provide a microscopic explanation for this phenomenon: under prolonged thermal stress, the polyp expels a large proportion of its zooxanthellae, thereby losing its main source of nutrients and its colour, and becoming weakened and vulnerable.

Finally, these lessons can be put into context through an introduction or discussion on the importance of preserving coral reefs (Box 2).

Box 2. The importance of coral reefs for marine biodiversity and humans

Coral reefs are extremely diverse ecosystems, home to a wide variety of corals and numerous other marine organisms that interact within complex ecological networks. Although they cover less than 1 per cent of the ocean floor, they are home to at least 25 per cent of marine species and thus play a major role in maintaining marine biodiversity [8].

Reefs also represent a vital resource for many coastal communities. They contribute to food security and livelihoods, particularly through fishing and tourism, whilst providing natural protection for coastlines against waves, storms and erosion [9].

5. Modeling and didactic choices

The threshold values presented in Section 1 are purely indicative for the three genera in the Pacific [10], as all three genera exhibit variability both between sites and between individuals [1]. We have chosen not to include this variability in the simulation so that students can draw their own conclusions about the genera. The threshold values are also considered to be independent of temperature, as the effect of the duration of a heatwave is dominant [10].

Microscopic view

We have chosen to depict a single coral polyp of the genus *Acropora*, in order to focus on the events occurring at the microscopic level following a rise in sea temperature. A single tentacle is shown in cross-section, to illustrate the location of the zooxanthellae within the host animal's tissue.

Within the tentacles, the zooxanthellae are housed in the polyp's gastrodermal cells, also known as symbiocytes (cells that host the symbiosis). The polyps appear brownish due to the photosynthetic pigments in their zooxanthellae¹.

Macroscopic view

Three genera of corals are used to illustrate the potential coexistence between them and the importance of biodiversity within a single coral reef. We chose to identify the specimens at genus level (*Acropora*, *Pocillopora* and *Porites*) because species diversity varies considerably between different reef sites [1]. These three genera of coral are among the most frequently observed in tropical Pacific reef communities, with variations in composition depending on the sites studied [12].

¹ It has long been suggested that the fluorescent pigments in the polyp, in low-light habitats, could enhance coral photosynthesis by modulating the spectrum and intensity of the light reaching the symbiotic cells [11].

References

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